No. of Printed Pages : 4

MCS–024

MASTER OF COMPUTER APPLICATIONS (MCA) (REVISED)/ BACHELOR OF COMPUTER APPLICATIONS (BCA) (REVISED)

Term-End Examination

June, 2024

MCS-024 : OBJECT ORIENTED TECHNOLOGIES AND JAVA PROGRAMMING

Time : 3 Hours *Maximum Marks : 100*

Weightage : 75%

Note : *Question No.* ***1*** *is compulsory. Attempt any* ***three*** *questions from the rest.*

---

1. (a) What is 'this' keyword ? Explain its usage with an example. 5

   (b) Compare AWT and Swing components. 5

   (c) Write a Java program to find the factorial of a number given by the user. Support your code with suitable comments. 5

(d) Discuss the life cycle of Applet, with the help of a suitable block diagram. 5

(e) What is a Stream socket ? How is it different from a Datagram socket ? 5

(f) What is a literal in Java ? How many types of literals are there in Java ? 5

(g) Explain, how threads are created by implementing Runnable Interface. What is Thread priority ? 5

(h) Briefly discuss the term 'Garbage collection'. Give the advantages and disadvantages of Garbage collection. 5

2. (a) Explain function overloading with the help of an example code in Java. 5

(b) Explain dynamic building. Also give advantages of dynamic binding. 5

(c) Why main method in Java is static ? Justify your answer. 5

(d) Compare StringBuffer object with String object. When should StringBuffer object be preferred over string object ? 5

3. (a) Explain each component of the statement given below :

'public static void main (string args[ ])'.

Do we use this statement, while writing code for Applet or Servlet. Justify your answer. 8

(b) What is a classpath ? Explain the utility of classpath with suitable example. 5

(c) Explain the Java RMI architecture with the help of a diagram. 7

4. (a) Write a Java program that reads the content of a text file and counts the number of white spaces, number of characters, number of words, and number of full stops in the content. Support your code with comments, to describe your logic. 10

(b) What is Multithreading ? How does process differ from thread ? Explain the steps for creating thread in Java. Also explain the predefined thread priorities in Java. 10

5. Write short notes on the following : 4×5=20

(a) Transient and volatile modifiers

(b) Object serialization

(c) Exception handling

(d) Abstract class

(e) Exception handling in Java

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MCS-024

MASTER OF COMPUTER APPLICATIONS/ BACHELOR OF COMPUTER APPLICATIONS

[MCA (REVISED)/BCA (REVISED)]

Term-End Examination

December, 2023

MCS–024 : OBJECT ORIENTED TECHNOLOGIES AND JAVA PROGRAMMING

Time : 3 Hours *Maximum Marks : 100*

Weightage : 75%

Note : *Question No.* ***1*** *is compulsory. Attempt any* ***three*** *questions from the rest.*

1. (a) What is a class ? How does it accomplish data hiding ? Explain with an example. 5

   (b) Differentiate between AWT and Swing components. Give suitable example for each. 5

P. T. O.

(c) Explain the relationship between inheritance and polymorphism. Give example of each. 5

(d) What is Java Bean ? Briefly discuss the features of Java Bean. 5

(e) Explain why main( ) method is not included in writing a Java applet program using an applet coding. 5

(f) Compare structured programming and object oriented programming. 5

(g) Discuss the utility of final and finalize keywords in Java. Give example code for each. 5

(h) How does datagram socket differ from stream socket ? 5

2. (a) Write a Java applet to display. “Java is an object oriented programming language” in red color. Support your code with suitable comments. 5

(b) What is a layout manager ? Explain the flow layout and border layout. Also explain how a layout is set. 8

(c) What are packages in Java ? How do we create a package in Java ? What are the steps to add Classes and Interfaces in a package ? 7

3. (a) Differentiate between thread and process. Explain Java thread model using thread life cycle. 7

(b) Compare Applet and Servlet. Write the steps to incorporate applet in a webpage. 8

(c) Explain method overriding in Java using a program in Java. 5

4. (a) Write a Java program for writing files in "C:\javafiles\" location. Support your program with suitable comments. 7

(b) Write the steps of JDBC in establishing a connection for creating dynamic website for INSERT/UPDATE the attendance record for students of a college. 7

(c) What is synchronization ? Explain how methods are synchronized in Java with the help of an example. 6

5. Write short notes on the following : 4×5=20

(a) Object serialization

(b) Liberals and its types

(c) Servlet life cycle

(d) Garbage collection

(e) Role of cookies in session handling

No. of Printed Pages : 4

MCS-024

MASTER OF COMPUER APPLICATIONS (MCA) (REVISED)/ BACHELOR OF COMPUTER APPLICATIONS B.C.A. (REVISED)

Term-End Examination

June, 2023

MCS-024 : OBJECT ORIENTED TECHNOLOGIES AND JAVA PROGRAMMING

Time : 3 Hours *Maximum Marks : 100*

Weightage : 75%

Note : *Question* ***No. 1*** *is compulsory. Attempt any* ***three*** *questions from the rest.*

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1. (a) Explain the basic features of object oriented programming language. 5

   (b) Why is Java called machine independent language ? Explain the functionality of JVM. 5

(c) Differentiate between constructor and method, give example for both. 5

(d) What is an abstract class ? Explain the use of abstract class with an example. 5

(e) What are cookies and session variables ? Briefly discuss the utility of both. 5

(f) Explain the different steps in the life cycle of an applet. 5

(g) Discuss the relationship between data abstraction and encapsulation. 5

(h) Explain the usage of container in Java Applet and Servlet. 5

2. (a) Explain how exception handling is performed in Java. Briefly discuss the concept of checked exception and unchecked exception, with an example of each. 10

(b) Briefly discuss the concept of listener in Java. Write a program in java to implement mouse motion listener, support your program with suitable comments. 10

3. (a) What is string class ? How is it different from String Buffer class ? Write a java program to find the length of a given string. 7

(b) What is servlet ? Explain how session handling is performed in servlet programming. 7

(c) What is an Interface ? Write a java program to show how a class implements two interfaces. 6

4. (a) What is RMI in Java ? Write the steps to create stub and skeleton. 7

(b) Write a java program to create a volume class. Derive cube and sphere classes from it. Define constructor for these three classes. 7

(c) What is Synchronization ? Explain how methods are synchronized in Java, with the help of an example. 6

5. Write short notes on the following : 5×4=20

(a) Dynamic Binding

(b) Stream Tokenizer

(c) JDBC Drivers

(d) Function Overloading

No. of Printed Pages : 4

MCS-024

M. C. A. (REVISED)/B.C.A. (REVISED)

Term-End Examination

December, 2022

MCS-024 : OBJECT ORIENTED TECHNOLOGIES AND JAVA PROGRAMMING

Time : 3 Hours *Maximum Marks : 100*

Weightage : 75%

Note : *Question* ***No. 1*** *is compulsory. Attempt any* ***three*** *questions from the rest of the questions.*

---

1. (a) What is method overloading ? Explain with suitable example. 5

   (b) Explain the relationship between inheritance and polymorphism through an example. 5

   (c) Distinguish between 'Final', 'Finally' and 'Finalize' keywords. Give example for each. 6

(d) Explain container class, and discuss its significance in Java GUI programming. 4

(e) Explain each component of the statement 'public static void main (string args[ ])'. 5

(f) What is Java applet ? How is it different from Java application program ? 5

(g) Explain the Java RMI architecture with the help of a diagram. 5

(h) What is Java Bean ? Give advantages of java beans in programming. 5

2. (a) Explain the usage of Grid Layout and Grid Bag Layout in a Java Applet. 5

(b) How String Class differs from String Buffer Class ? Explain with suitable example. 5

(c) What is an applet ? Discuss all the phases of applet life cycle. 5

(d) Explain GET and POST methods of servlet programming, with the help of suitable code. 5

3. (a) Write a java program to create a file named 'ignou-txt' and write 'Hello IGNOU' in it. 6

(b) Write a java program for 'writing output on console' using PrintWriter( ) method. Give suitable comments for readability of your program. 6

(c) What is exception in Java ? What are the basic causes of occurrence of an exception ? Write a program in Java, to show the concept of exception handling. Support your program with suitable comments. 8

4. (a) What is inheritance ? How is inheritance related to interface ? Explain, how multiple inheritance can be implemented using interface in Java. Give suitable example. 10

(b) Explain the steps of JDBC in establishing a connection for creating a dynamic website for INSERT/UPDATE the attendance record for employees of an organization. 10

5. (a) Write a Java program which takes two 3×3 matrices as input and find sum of the two matrices ? Define constructor for initializing the matrix object. Give suitable comments for readability of your code. 10

(b) Write a Java program to print factorial of a number entered by user. 5

(c) Compare object oriented programming and structured programming. Give advantages of object oriented progamming over the structured programming. 5

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MCS-024

MCA (Revised) / BCA (Revised)

Term-End Examination

June, 2022

MCS-024 : OBJECT ORIENTED TECHNOLOGIES AND JAVA PROGRAMMING

Time : 3 hours *Maximum Marks : 100*

(Weightage : 75%)

Note : *Question no.* ***1*** *is* ***compulsory****. Answer any* ***three*** *questions from the rest.*

---

1. (a) What is an abstract class ? How is it used to implement polymorphism in Java ? Give suitable example in support of your answer. *5*

(b) What is an applet ? Why is main( ) not included in writing a Java applet program ? *5*

(c) Explain the process of inter-thread communication in Java, with the help of a suitable example. *5*

(d) Differentiate between throw with throws statement. Give example code for each. *5*

(e) What is classpath ? Explain the utility of classpath, with the help of a suitable example. *5*

(f) Compare private and protected access specifiers. Give suitable example for each. *5*

(g) Differentiate between Method-overloading and Method-overriding, with the help of a suitable example. *5*

(h) What is layout manager ? Explain any two layouts in Java. *5*

2. (a) What are proxy servers ? Explain the usage of anonymous proxy servers in designing mailing applications. *6*

(b) How does Servlet differ from Applet ? Discuss all the phases of a servlet lifecycle. *7*

(c) Write a program in Java to copy the text content of one file into another file. Support your program with suitable comments. *7*

3. (a) What is StringBuffer class ? Write a program in Java to append a given string to a StringBuffer object. *6*

(b) Write a program to pass parameters to an Applet using a web page. Also write the phases of an Applet lifecycle. *6*

(c) What is an event in Java ? How does Java handle events ? Write a program in Java to capture any event generated by keyboard. *8*

4. (a) Write a Java program to create a Teacher class and derive PGT, TGT and PRT classes from Teacher class. Define appropriate construction for all the classes. Also define a method to display information of Teacher. Make necessary assumptions as required. *10*

(b) Write a Java program to set up JDBC and execute an SQL statement on a database table 'BOOK', with fields BOOK_ISBN, BOOK_AUTH, BOOK_PRICE, BOOK_PUBL. The SQL statement should find the BOOK_AUTH and BOOK_PRICE of the BOOK having ISBN = 500123. *10*

5. (a) What is an Exception ? Briefly explain the causes of Exceptions. Describe how multiple exceptions are caught in Java, with the help of a suitable program code. *10*

(b) Write a Java program to print Fibonacci series, up to the nth term entered by the user. Support your program with suitable comments. *5*

(c) Explain the Java RMI architecture, with the help of a diagram. *5*

No. of Printed Pages : 3

MCS-024

MCA (Revised) / BCA (Revised)

Term-End Examination

December, 2021

MCS-024 : OBJECT ORIENTED TECHNOLOGIES AND JAVA PROGRAMMING

Time : 3 hours *Maximum Marks : 100*

(Weightage : 75%)

Note : *Question no.* ***1*** *is* ***compulsory****. Answer any* ***three*** *questions from the rest.*

---

1. (a) Compare data abstraction and data hiding. Give an example of each. *5*

(b) Compare Inheritance and Polymorphism. Give an example of each. *5*

(c) Why is Java called machine independent and architecture neutral ? What is the role of JVM (Java Virtual Machine) ? *5*

(d) Write a JAVA program to find the factorial of a given number. Define appropriate class and methods. *5*

(e) What is an exception ? Write any three actions that can be taken after an exception occurs in a program. *5*

(f) What are threads ? Explain how threads are created by implementing Runnable Interface. *5*

(g) What is serialization ? Differentiate between Transient and Volatile keyword. *5*

(h) Discuss the architecture of Applet, with the help of a suitable block diagram. *5*

2. (a) Briefly discuss the methods used by Applet during their life cycle. *5*

(b) What are proxy servers ? When should you use anonymous proxy servers ? *5*

(c) What is RMI ? Which protocol does it use ? Discuss the security problem in RMI. *5*

(d) What is session tracking ? Discuss the role of cookies in session tracking. *5*

3. (a) What is Java Bean ? Briefly discuss the features of Java Beans. How does Java Bean differ from an instance of a normal Java class ? *6*

(b) Explain method overloading and method overriding. Give suitable example for each. *8*

(c) What is Synchronization ? Why is it important ? Explain the use of synchronized method in multithreading. *6*

4. (a) What is the difference between & operator and the && operator ? Give suitable example for each. *5*

(b) What is a constructor ? Explain the use of constructor with the help of a program. *5*

(c) What is finalize( ) method ? Write the properties of finalize( ) method. Write the declaration statement for finalize( ) method. *5*

(d) Compare checked exceptions with unchecked exceptions. Give an example for each. *5*

5. Write short notes on the following : *4×5=20*

(a) Methods Used for Interthread Communication

(b) Object Serialization and its Working

(c) Package in Java

(d) AWT and Swing

No. of Printed Pages : 3

MCS-024

MCA (Revised) / BCA (Revised)

Term-End Examination

June, 2021

MCS-024 : OBJECT ORIENTED TECHNOLOGIES AND JAVA PROGRAMMING

Time : 3 hours *Maximum Marks : 100*

(Weightage 75%)

Note : *Question no.* ***1*** *is* ***compulsory****. Attempt any* ***three*** *questions from the rest.*

---

1. (a) Compare Object Oriented and Procedure Oriented approaches. Give salient features of both. *5*

(b) What is Dynamic Binding ? Explain the advantage of dynamic binding. *5*

(c) Explain Function Overloading with the help of suitable example. *5*

(d) Explain why the main method in Java is always static. *5*

(e) What is an Interface ? Write a program to show how a class implements two interfaces. *5*

(f) What is Object Serialization ? Briefly discuss the working of Object Serialization. *5*

(g) Compare StringBuffer Object with String Object. When should StringBuffer Object be preferred over String Object ? *5*

(h) What is Stream Tokenizer ? Discuss with suitable example. *5*

2. (a) Discuss the life cycle of applet, with the help of a suitable block diagram. *5*

(b) What is a Stream Socket ? How is it different from Datagram Socket ? *5*

(c) Briefly discuss the different kinds of drivers used in JDBC. *5*

(d) How do Servlets differ from Applets ? Briefly discuss the Servlet Life Cycle. *5*

3. (a) What is RMI ? Briefly discuss the RMI object hierarchy. Discuss how distributed applications are created using RMI. *10*

(b) Write a JAVA program that overloads the function 'SUM' that computes the sum of two integers and the sum of two float valves. *5*

(c) Compare doGet() and doPost() method. *5*

4. (a) What is a Literal ? How many types of literals are there in Java ? *5*

(b) Briefly discuss the term Garbage Collection. Give advantages and disadvantages of Garbage Collection. *5*

(c) What are uses of final keyword in Java ? *5*

(d) Explain how threads are created by implementing Runnable Interface. What is thread priority ? *5*

5. Write short notes on the following : *4×5=20*

(a) Transient and Volatile Modifiers

(b) Random Access File and its Methods

(c) TCP/IP Sockets

(d) Multidimensional Array in JAVA

No. of Printed Pages : 4

MCS-024

M. C. A. (REVISED)/B.C.A. (REVISED)

Term-End Examination

December, 2020

MCS-024 : OBJECT ORIENTED TECHNOLOGIES AND JAVA PROGRAMMING

Time : 3 Hours *Maximum Marks : 100*

Weightage : 75%

Note : *Question* ***No. 1*** *is compulsory. Attempt any* ***three*** *questions from the rest of the questions.*

---

1. (a) What is Multithreading ? How is multithreading supported in Java ? 5

   (b) What is explicit casting ? Explain with suitable example. 5

   (c) What is the difference between declaring a variable and defining a variable ? Explain with the help of an example. 5

(d) Explain the following and their usage using suitable examples : 5

(i) Extends

(ii) Implements

(e) Write a java program to convert a given string to uppercase and also display the length of the string. 5

(f) What is datagram socket ? Explain the use of datagram socket with the help of a program in java. 5

(g) Write a java program to draw a triangle. 5

(h) What is function overloading ? Explain it with the help of a program, written in java. 5

2. (a) How does java application differ from java applet ? Do we write "public static void main(string args [ ]) in applet ? Justify. 5

(b) Write a java program to copy the text content of one file into another file. Support your program with suitable comments. 7

(c) What is the relation between interface and inheritance ? Explain how multiple inheritance can be implemented in Java using interface. 8

3. (a) Explain the life cycle of servlet, along with a program that creates a simple servlet generating HTML to write "welcome students". 10

(b) What is an exception ? What is runtime exception ? Write a java program to search element in a given array with provision of exception handling. 10

4. (a) How java implements the model of interprocess synchronization using threads ? Explain with the help of a program. 10

(b) What is Inheritance ? What are the advantages of inheritance ? Write a java program to demonstrate implementation of inheritance. 10

5. Write short notes on the following : $5 \times 4 = 20$

(a) Cookies and their role in session handling

(b) Layout manager and its role

(c) Buffered stream classes and their utility

(d) Uses of super keyword

(e) Garbage collection

No. of Printed Pages : 4

MCS-024

MCA (Revised) / BCA (Revised)

Term-End Examination

MCS-024 : OBJECT ORIENTED TECHNOLOGIES AND JAVA PROGRAMMING

Time : 3 Hours] *[Maximum Marks : 100*

(Weightage 75%)

Note: Question number 1 is compulsory. Answer any three questions from the rest of the questions.

---

1. (a) What do you understand by "method overloading"? How it is different from "method overriding"? Give suitable example for each. 5

   (b) What is the purpose of setting class path? How it supports the execution of Java program? Write the steps, performed to set the class path. 5

   (c) What do you mean by "garbage collection" in Java? Give advantages and disadvantages of garbage collection. 5

(d) What is interface in Java? Write a java program to exhibit, how to define and implement interface. 5

(e) What are the uses of "this" keyword in java? Explain with the help of an example. 5

(f) What is byte code? Explain the difference between compiled code of Java and compiled code of C. 5

(g) Write a program in Java to convert the string "IGNOU" in lower case i.e. "ignou". Give suitable comments in your code, to increase the readability. 5

(h) What is an abstract class? How it is used to impliment polymorphism in Java? 5

2. (a) Explain the life cycle of applet, with suitable diagram. 5

(b) Write a program in Java that reads text from the key board and writes it to a ".txt" file. Incorporate suitable comments in your program, to improve the readability of your logic. 7

(c) Explain "<<", ">>" and "&" bitwise operators in Java. Write a Java program to perform "<<" bitwise operator on a given number. 8

3. (a) What is certainer (in context of java)? Explain the significance of container in Java GUI program. 5

(b) What is stream tokenizer in Java? What is its significance ? Discuss the instance variables defined in streamtokenizer? Give an example to show the use of streamtokenizer. 7

(c) Write a client & server programs in Java to show the TCP connection establishment and data transfer. 8

4. (a) Write a JAVA program to draw a color filled line and a color filled rectangle. 5

(b) Explain the term RMI in Java. Briefly discuss the steps involved in creation of stub and skeleton. 7

(c) Discuss the concept of multithreading in Java. What are the advantages of multithreading? Explain the interthread communication with the help of an example. 8

5. (a) What do you understand by "exception", in context of Java? How it is handled, explain with the help of suitable example code in Java? 5

(b) What is a layout manager? Briefly discuss the role of layout manager in Java. Explain the flow layout and grid layout with the help of an example for each. 7

(c) Differentiate between <u>any two</u> of the following: 8

(i) String class and stringBuffer class

(ii) Throw statement and throws statement

(iii) Final and Finally

—x—

No. of Printed Pages : 4

MCS-024

MCA (Revised) / BCA (Revised)

Term-End Examination, 2019

MCS-024 : OBJECT ORIENTED TECHNOLOGIES AND JAVA PROGRAMMING

Time : 3 Hours] [Maximum Marks : 100

(Weightage 75%)

Note : Question **no. 1** is **compulsory** and carries 40 marks. Attempt **any three** questions from the rest.

1. (a) How String Buffer differs from String ? Write a program in Java, which takes your name as input and print it in upper case. [5]

(b) How does Java handle events ? Write a program in Java to capture an event generated by keyboard. [5]

(c) What are the uses of "this" keyword in Java ? Explain with the help of an example. What is an applet ? [5]

(d) Write a program to generate a circle in an applet ? [5]

(e) What do you understand by session tracking in Java ? Discuss any technique, used for session tracking. [5]

(f) Differentiate between superclass and subclass. Write a program in Java to show the concept of reusability. [5]

(g) What is exception in Java ? How exception handling is performed, demonstrate with an example ? [5]

(h) Explain constructor overloading with suitable example. [5]

2. (a) What is checked exception in java ? How it is different from unchecked exception ? Explain briefly. [5]

(b) Write a Java program to find whether a given element is present in a integer array of size ten of not. [5]

(c) What is a layout manager ? Discuss the Flow layout and grid layout, give suitable example for each. [10]

3. (a) Briefly explain the role of following classes in Java Network programming : [5]

(i) Socket

(ii) Datagram Packet

(b) What is the difference between a process and a thread ? Mention any three methods of thread class and describe them briefly. [8]

(c) What is an event ? What are components of an event ? Differentiate between semantic event and low-level event. [7]

4. (a) Write a Java program to create shape class. Overload method area ( ) to calculate area of circle and rectangle. Define appropriate constructors in the program. [6]

(b) What are Input stream and output stream classes in java ? List and explain any two methods of each class. [5]

(c) Explain how garbage collection takes place in java, also list advantages of garbage collection. [4]

(d) What is an interface ? Write a java program to show, how a class implements two interface. [5]

5. Write short notes on the following : [20]

(i) Java Beams and its advantages

(ii) Class and object

(iii) Applet Architecture

(iv) Access specifiers

(v) Bitwise operators in java

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No. of Printed Pages : 3

MCS-024

MCA (Revised) / BCA (Revised)

Term-End Examination, 2019

MCS-024 : OBJECT ORIENTED TECHNOLOGIES AND JAVA PROGRAMMING

Time : Three Hours] [Maximum Marks : 100 (Weightage 75%

Note : Question No. **1** is **Compulsory**. Attempt **any three** questions from the rest.

1. (a) Write a java program to read the contents of binary file and write it on the standard output. [10]

   (b) What is purpose of Serialization of an object ? How does the volatile modifier to a data type affect serialization ? [5]

   (c) Justify "Java is secure and architectural neutral language". [3]

   (d) What are cookies and session objects ? Where do we use them ? [5]

   (e) Explain the concept of inheritance and polymorphism, with an example of each. [7]

(f) Differentiate between AWT and Swing Components. [5]

(g) What is the difference between notify( ) and notify All( ) method. [5]

2. (a) What is exception ? What are causes of exception ? What can be done once an exception is caught ? Explain use of finally clause in exceptions handling, with the help of an example. [10]

(b) Differentiate between following : [10]

(i) Public member and private member

(ii) Overloading and Overriding

(iii) Java application and Applet

(iv) FOR statement and WHILE statement

3. (a) What is Multithreading ? What is the advantage of multithreading in Java ? [5]

(b) What are Interfaces in Java ? What do you understand by 'implementing interfaces' ? Explain with the help of an example. [7]

(c) What is JDBC ? What are the characteristics of JDBC ? Write a program to demonstrate how JDBC connection is established. [8]

4. (a) What is String in Java ? Explain any three constructors of String Class. Write a program in Java to find length of a given string and display the string in upper case. [10]

(b) What is servlet ? How servlet differs from Applet? Explain the lifecycle of Applet. [10]

5. (a) Write a Java program to create student class with data member student_name, roll-number and Marks. Define appropriate constructor in the program. Create array of ten student objects. Also display the name and rollnumber of the student with highest marks (in the array created). [10]

(b) What is RMI ? Explain RMI architecture with suitable diagram. [5]

(c) What are Java Beans ? Discuss the utility of Java Beans. Explain the features of Java Beans. [5]

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No. of Printed Pages : 3

MCS-024

MCA (Revised) / BCA (Revised)

Term-End Examination

December, 2018

08553

MCS-024 : OBJECT ORIENTED TECHNOLOGIES AND JAVA PROGRAMMING

Time : 3 hours *Maximum Marks : 100*

(Weightage : 75%)

Note : *Question no. **1** is **compulsory**. Attempt any **three** questions from the rest.*

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1. (a) What do you understand by 'this' keyword ? Explain its usage with an example. *5*

(b) What is an event in Java ? Describe different components of an event. *5*

(c) What is an interface ? Discuss the utility of interface with suitable example. *5*

(d) What are cookies ? Discuss the role of cookies in session tracking. *5*

(e) Compare AWT and Swing components. *5*

(f) Write a Java program, to find factorial of a given number. *5*

(g) What is a layout manager ? Explain Border layout with an example. *5*

(h) Explain the Java RMI architecture with the help of a diagram. *5*

2. (a) Write a Java program that reads the content of a text file and counts the number of white spaces, number of characters, number of words and number of full stops in the content. Support your code with comments, to describe your logic. *10*

(b) What is multithreading ? Explain the steps for creating thread in Java. Also explain the predefined thread priorities in Java. *10*

3. (a) What are class and objects ? Explain how an object is created in Java ? Also explain the role of constructor in Java using Java code. *10*

(b) What is Polymorphism ? Explain the difference between method overloading and method overriding with the help of example. *10*

4. (a) Explain the life cycle of Applet. List the methods involved in it. *10*

(b) Write a Java program which takes two 3×3 matrices as input and finds the sum of the two matrices. Define constructor for initialization of matrix objects. *10*

5. Write short notes on the following (Give example for each) : *4×5=20*

(a) Message Passing

(b) Java Beans

(c) Strings in Java

(d) Package

No. of Printed Pages : 3

MCS-024

MCA (Revised) / BCA (Revised)

Term-End Examination

June, 2018

07005

MCS-024 : OBJECT ORIENTED TECHNOLOGIES AND JAVA PROGRAMMING

Time : 3 hours *Maximum Marks : 100*

(Weightage 75%)

Note : *Question no. **1** is **compulsory**. Attempt any **three** questions from the rest of the questions.*

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1. (a) What is method overloading ? Explain with suitable example. *5*

(b) What is an abstract class ? How is it used to implement polymorphism in Java ? Give suitable example in support of your answer. *7*

(c) What is JVM ? Explain advantages of JVM. *3*

(d) Differentiate between "Final", "Finally" and "Finalize" statements. *6*

(e) Compare throw statement in Java with throws statement. *4*

(f) Explain container class in Java. Give its significance for Java GUI programming. *5*

(g) What is public access specifier ? How is it different from private access specifier ? Explain with example. *5*

(h) What is an applet ? Explain various ways to execute an applet. *5*

2. (a) What is servlet life cycle ? Explain GET and POST methods in servlet programming. *10*

(b) What is import ? Explain the need of importing a package in Java. *5*

(c) What is a classpath ? Discuss the utility of classpath with suitable example. *5*

3. (a) What is inheritance ? How is inheritance related to interface ? Explain, how multiple inheritance can be implemented using interface in Java. Give suitable example. *10*

(b) Write a Java program to create a Teacher class and derive Professor/ Associate_Professor/Assistant_Professor class from Teacher class. Define appropriate constructor for all the classes. Also define a method to display information of Teacher. Make necessary assumptions as required. *10*

4. (a) What are the features of Object Oriented Programming ? Explain encapsulation and its advantages, with an example. *6*

(b) How is String class different from StringBuffer class ? Write a program in Java to find the length of a given string. *7*

(c) Write a program in Java, to copy the text content of one file into another file. *7*

5. (a) Explain each component of the statement "public static void main (string args [ ])". *6*

(b) What is an event in Java ? How does Java handle events ? Write a program in Java, to capture any event generated by a keyboard. *7*

(c) What is exception in Java ? What are the basic causes of occurrence of exception ? Write a program in Java, to exhibit the concept of exception handling. Support your program with suitable comments. *7*

No. of Printed Pages : 5

MCS-024

MCA (Revised) / BCA (Revised)

Term-End Examination

December, 2017

04950

MCS-024 : OBJECT ORIENTED TECHNOLOGIES AND JAVA PROGRAMMING

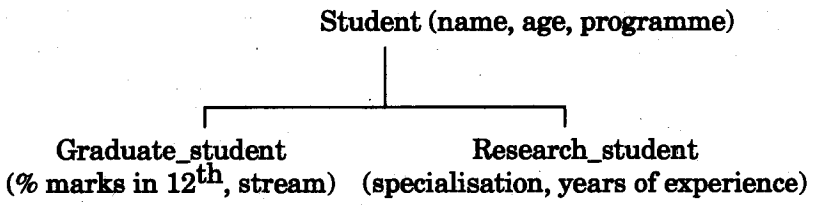
Time : 3 hours *Maximum Marks : 100*

(Weightage 75%)

Note : *Question no. 1 is* ***compulsory*** *and carries 40 marks. Attempt any* ***three*** *questions from the rest of the questions.*

---

1. (a) Consider the following class hierarchy :



A university has two types of students — graduate students and research students. The University maintains the record of name, age and programme of every student. For graduate

students, additional information like percentage of marks and stream, like science, commerce, etc. is recorded; whereas for research students, additionally, specialisation and years of working experience, if any, is recorded. Each class has a constructor. The constructor of subclasses makes a call to constructor of the superclass. Assume that every constructor has the same number of parameters as the number of instance variables. In addition, every subclass has a method that may update the instance variable values of that subclass. All the classes have a function display_student_info( ), the subclasses must override this method of the base class. Every student is either a graduate student or a research student.

Perform the following tasks for the description given above using Java :

(i) Create the three classes with proper instance variables and methods, with suitable inheritance. *5*

(ii) Create at least one parameterised constructor for each class. *3*

(iii) Implement the display_student_info( ) method in each class. *3*

(iv) Create an appropriate main method with at least two objects of each subclass. Can you create an object of the superclass ? Justify your answer. *3*

(v) Write the code in main that uses objects and overriding to show polymorphism. *3*

(b) What is multithreading ? What are the advantages of multithreading ? What is a main thread in the context of Java ? *5*

(c) Write a Java program that accepts the input from the keyboard and writes it to a text file. *5*

(d) Explain how an event is handled in Java with the help of an example program. *6*

(e) Assume that a database named "Student" exists with attributes student_ID, student_name and programme. Write a Java program segment which will execute a SELECT query and display the resultant records. You need not write the connection command. *4*

(f) What is Bytecode in Java ? What are the advantages of using bytecode in Java ? *3*

2. (a) List the salient features of object oriented programming approach that distinguishes it from the procedural programming. *4*

(b) Explain different forms of inheritance with the help of diagrams. What are the advantages of using inheritance ? *8*

(c) Explain the following with the help of an example of each : *8*

(i) Dynamic Initialization

(ii) Operator Precedence

(iii) Switch Statement

(iv) Array Initialization

3. (a) What is Static Method in Java ? Explain with the help of an example. *4*

(b) Explain with the help of an example program, how objects can be passed as parameters in Java. *5*

(c) What is a Package in Java ? How are they related to CLASSPATH ? Explain with the help of an example program. *6*

(d) What is an Interface in Java ? How are they different from ABSTRACT classes ? *5*

4. (a) Explain with the help of an example program, how interthread communication is performed in Java using wait( ) and notify( ) and other methods. *6*

(b) How is Character class in Java different to String class ? Explain how you will compare two objects of String class. Also write a program that converts lowercase characters to uppercase characters of a string. *8*

(c) What is the purpose of the following Stream classes ? *6*

(i) PrintStream

(ii) RandomAccessFile

(iii) ByteArrayInputStream

(iv) FilterOutputStream

5. Explain the following with the help of a/an diagram/example/program, if needed : *20*

(a) Paint( ) Method of Applet

(b) HTML Applet Tag

(c) Button

(d) Checkbox Group

(e) Grid Layout (No need of writing program)

(f) Container

(g) Uses of RMI

(h) POST Method

No. of Printed Pages : 4

MCS-024

MCA (Revised) / BCA (Revised)

Term-End Examination

10861 **June, 2017**

MCS-024 : OBJECT ORIENTED TECHNOLOGIES AND JAVA PROGRAMMING

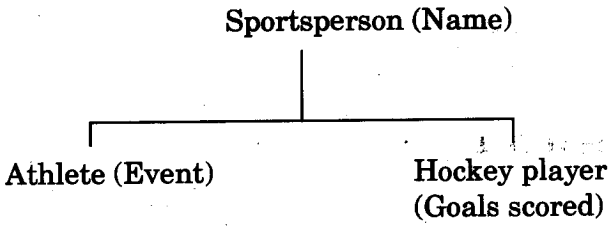
Time : 3 hours *Maximum Marks : 100*

(Weightage 75%)

Note : *Question no. 1 is* ***compulsory*** *and carries 40 marks. Attempt any* ***three*** *questions from the rest.*

1. (a) An employee works in a particular department of an organisation. Every employee has an employee number, name and draws a particular salary. Every department has a name and a head of department. The head of department is an employee. Every year a new head of department takes over. Also, every year an employee is given an annual salary enhancement. Identify and design the classes for the above description with suitable instance variables and methods. The classes should be such that they implement information hiding. You must give logic in support of your design. Also create two objects of each class. *6*

(b) Consider the following class hierarchy :



In this hierarchy, you can assume that a sportsperson can either be an athlete or a hockey player. Every sportsperson has a unique name. An athlete is characterised by the event in which he/she participates; whereas a hockey player is characterised by the number of goals scored by him/her. Perform the following tasks using Java :

(i) Create the class hierarchy with suitable instance variables and methods. *3*

(ii) Create a suitable constructor for each class. *3*

(iii) Create a method named display_all_info with suitable parameters. This method should display all the information about the object of a class. *3*

(iv) Write the main method that demonstrates polymorphism. *4*

(c) A text file stored on the disk is to be opened for reading operations. What are the different checks that should be performed on this file ? Write the portion of the code in Java to perform these checks. *4*

(d) Explain the thread model with the help of a diagram. *6*

(e) What is a layout manager ? Explain the flow layout and grid layout with the help of an example of each. *7*

(f) What is an event in the context of Java ? Explain the semantic event with the help of an example. *4*

2. (a) Differentiate between the following : *12*

(i) Procedural paradigm versus Object-oriented paradigm

(ii) While versus For statements of Java

(iii) Function overloading versus Method overriding

(iv) Java application versus Java applet

(b) Explain the importance/uses of the following with the help of an example each : *8*

(i) Bitwise operators

(ii) Operators precedence

(iii) Array initializer

(iv) Byte code

3. (a) Define the term exception in the context of Java. What are the causes of exception ? What are the actions that may need to be performed if an exception is caught ? Explain the uses of try and catch in Java with the help of an example program segment. *10*

(b) What is an Interface ? What is meant by 'implementing interfaces' ? Explain with the help of an example. *5*

(c) What are the uses of "this" keyword in Java ? Explain with the help of an example. *5*

4. (a) What are the advantages of multithreading in Java ? Explain the interthread communication with the help of an example. *8*

(b) Write a program in Java that converts a string 127 into equivalent integer value and prints it. *6*

(c) What are the uses of the following in the context of I/O in Java ? *6*

(i) File class

(ii) Buffered stream classes

5. Explain the following with the help of an example/diagram/program, if needed : *5×4=20*

(a) Creating a choice list in Java

(b) Using checkboxes in Java

(c) Drawing a rectangle in Java

(d) Servlet Life Cycle

(e) Executing SELECT statement in Java

No. of Printed Pages : 3

MCS-024

MCA (Revised) / BCA (Revised)

Term-End Examination

December, 2016

04995

MCS-024 : OBJECT ORIENTED TECHNOLOGIES AND JAVA PROGRAMMING

Time : 3 hours *Maximum Marks : 100*

(Weightage 75%)

Note : *Question no. 1 is* ***compulsory****. Attempt any* ***three*** *questions from the rest.*

---

1. (a) Explain the relationship between inheritance and polymorphism. *4*

(b) Explain why main() is not included in writing a Java applet program using an applet coding. *6*

(c) Explain method overloading using an example. *6*

(d) Explain the process of inter-thread communication. *6*

(e) Explain the usage of three assignment operators and three bitwise operators in Java programming. *6*

(f) Explain the usage of GridLayout and GridBagLayout in a Java Applet with diagram. *6*

(g) Which types of components are used in designing swing based GUI ? *6*

2. (a) Explain the usage of anonymous proxy servers in designing mailing application. *6*

(b) Explain the steps of using multidimensional arrays in a Java program. *6*

(c) Write a Java program for "writing output on console" using PrintWriter() method. *8*

3. (a) Describe all the steps used in writing a Java program for handling events. *6*

(b) What is the usage of RMI in calling methods from remote class ? *6*

(c) Write a Java program to show interprocess synchronization. *8*

4. (a) Explain multithreading using a program. *6*

(b) Explain the usage of lastIndexOf (int ch, int fromIndex) method in string operation. *6*

(c) Write a program for passing parameters to an Applet using a web page. *8*

5. (a) Explain the term BufferedStream using a Java program. *6*

(b) Write a Java program for writing files in "C:\javafiles\" location. *6*

(c) Explain the steps of JDBC in establishing a connection for creating a dynamic website for INSERT/UPDATE the attendance record for employees of an organization. *8*

No. of Printed Pages : 3

MCS-024

MCA (Revised) / BCA (Revised)
Term-End Examination
June, 2016

09646

MCS-024 : OBJECT ORIENTED TECHNOLOGIES AND JAVA PROGRAMMING

Time : 3 hours *Maximum Marks : 100*

(Weightage 75%)

Note : *Question no. **1** is **compulsory**. Attempt any **three** questions from the rest.*

---

1. (a) Explain the relationship between Data Abstraction and Encapsulation. *4*

(b) Explain the need of main( ) function in writing a Java program using an example program. *6*

(c) Explain function overloading using an example. *6*

(d) Explain Java Thread Model using Thread Life Cycle. *6*

(e) What is the usage of literals in a programming language ? Explain the usage of any six Java literals used in Java programming, with examples. *6*

(f) Explain the usage of Flow Layout and Card Layout in Java Applet with diagram. *6*

(g) Explain the usage of container in Java Applet and Servlet. *6*

2. (a) Explain the usage of reserved ports in net programming for FTP, TCP, etc. *6*

(b) Explain all steps of using arrays in a Java program. *6*

(c) Write a program for reading input from the console using StringBuffer( ). *8*

3. (a) Explain all the components used for handling an event in a Java program. *6*

(b) What are Java beans ? Explain their special features. *6*

(c) Write a Java program to show inter thread communication. *8*

4. (a) Explain method overriding using a program. *6*

(b) How does an index method help in string operations ? *6*

(c) Write the steps of incorporating Applet in a web page. *8*

5. (a) Explain the usage of Stream Tokenizer class. *6*

(b) Write a Java program to convert all the strings of input to upper case. The input string should be submitted using a textbox in a form of web page. *6*

(c) Write the steps of JDBC in establishing a connection for creating dynamic website. *8*

08874

No. of Printed Pages : 3

MCS-024

MCA (Revised) / BCA (Revised)
Term-End Examination
December, 2015

MCS-024 : OBJECT ORIENTED TECHNOLOGIES AND JAVA PROGRAMMING

Time : 3 hours *Maximum Marks : 100*

(Weightage 75%)

Note : *Question no. 1 is* ***compulsory****. Attempt any* ***three*** *questions from the rest.*

---

1. (a) Explain the basic features of Object Oriented Programming Language. *5*

(b) Why is Java called Machine Independent Language ? Explain the functionality of JVM. *5*

(c) What is the difference between constructor and method ? Explain with an example. *5*

(d) Explain the uses of keywords *final*, *finally* and *finalize* in Java. *6*

(e) What is an abstract class ? Explain the use of abstract class with an example. *6*

(f) What are the different boolean and bitwise operators in Java ? Explain in brief. *6*

(g) Write a Java Applet to display "Java is an Object Oriented Programming Language"; in red color. *7*

2. (a) Distinguish between the following terms with suitable example(s) : *4×3=12*

(i) Method Overloading and Overriding

(ii) Application and Applet

(iii) Get Method and Post Method in Servlet.

(iv) Readers / Writers and I/O Streams.

(b) What is listener ? Write a program to implement mouse motion listener. *8*

3. (a) What is encapsulation ? Explain its advantage with suitable example. *3*

(b) What will be the output of the following program : *5*

```
Class t extends Thread
{
  public void run ( )
  {
    System.out.println ("start");
    yield ( );
    resume ( );
    System.out.println ("restart");
    stop ( );
    resume ( );
    System.out.println ("Nothing");
  }
  public static void main (String a [ ])
  {
    t1 = new t ( );
    t1.start ( );
  }
}
```

(c) Explain the output stream class hierarchy in Java. *6*

(d) What is layout manager ? Explain the Flow Layout and Border Layout. Also explain how a layout is set. *6*

4. (a) What is a JavaBean ? Discuss its features in brief. *5*

(b) What is checked exception ? How is it different from unchecked exception ? Explain with an example. *7*

(c) What is inheritance ? Create a class Train and inherit class SuperfastTrain from it, define proper constructor for both of the classes. *8*

5. (a) Write a program in Java to implement socket programming using DataGram Class. *8*

(b) How do we design a package in Java ? What are the steps to add classes and interfaces in a package ? *6*

(c) What is synchronization ? Explain how methods are synchronized in Java, with the help of an example. *6*

No. of Printed Pages : 3

MCS-024

MCA (Revised) / BCA (Revised)

Term-End Examination

07823 **June, 2015**

MCS-024 : OBJECT ORIENTED TECHNOLOGIES AND JAVA PROGRAMMING

Time : 3 hours *Maximum Marks : 100*

(Weightage 75%)

Note :

*(i) Q. No. **1** is **compulsory**.*

*(ii) Attempt any **three** questions from the rest.*

1. (a) What is Class ? How does it accomplish Data Hiding ? Explain with an example. *5*

(b) State any three differences between AWT and Swing components. *6*

(c) Explain the concept of Inheritance and Polymorphism with an example of each. *8*

(d) What do you mean by serialization ? What is the common usage of serialization ? *5*

(e) What are cookies and session variables and where do we use them ? *5*

(f) Write a Java program to read text from keyboard and write it to a file called 'MyText.txt'. *6*

(g) Explain any five basic features of Java in brief. *5*

2. (a) Differentiate between the following terms with examples : *3×4=12*

(i) Structured Programming and Object Oriented Programming

(ii) Data Abstraction and Encapsulation

(iii) length and length( )

(iv) Class and Object

(b) What are the two methods to create threads ? Explain them with a suitable example. *8*

3. (a) What is RMI ? How can you create stub and skeleton ? Write the steps involved. *7*

(b) Write a program to read the content of the file 'MyDoc.doc' and display it on console. *6*

(c) What is exception ? Explain the basic causes of exception. Write a Java program to explain how exceptions are handled. *7*

4. (a) Write a Java program to create a Shape class. Derive Circle and Square classes from it. Define constructor for these three classes. *7*

(b) Explain the different steps in the life cycle of an applet. *5*

(c) What is a package ? Develop a Java package with simple queue and stack class. *8*

5. (a) What is Servlet ? Explain how session handling is performed in Servlet programming. *7*

(b) What is an Interface ? Write a Java program to show how a class implements two interfaces. *6*

(c) What is String class ? How is it different from StringBuffer class ? Write a Java program to find the length of a given string. *7*

No. of Printed Pages : 3

MCS-024

MCA (Revised) / BCA (Revised)
Term-End Examination
December, 2014

11620

MCS-024 : OBJECT ORIENTED TECHNOLOGIES AND JAVA PROGRAMMING

Time : 3 hours *Maximum Marks : 100*

(Weightage 75%)

Note :

*(i) Q. No. **1** is **compulsory**.*

*(ii) Attempt any **three** questions from the rest.*

(iii) Attempt all parts of a question together.

---

1. (a) Define the concept of inheritance. Explain the advantage of inheritance with the help of an example. *7*

(b) What is classpath ? Explain the use of classpath in Java programming. *4*

(c) What is multithreading ? How is thread synchronization implemented in Java ? Explain with an example. *8*

(d) What is an event in GUI programming ? Explain different components of an event. *6*

(e) What is inner class ? Explain whether inner class is secure and useful. Justify your answer. *6*

(f) What is *final* keyword ? Explain different uses of *final* in Java. *4*

(g) What is exception ? Explain difference between checked and unchecked exceptions with an example of each. *5*

2. (a) Differentiate the following, with the help of example : *4×3=12*

(i) Application and Applet

(ii) Structure Approach and Object Oriented Approach

(iii) String and String Buffer

(iv) Abstract Class and Interface

(b) Write a class *complex* to represent complex numbers, with suitable constructor and function to find the sum of two complex numbers. *8*

3. (a) What is *StreamTokenizer* ? Explain the different instance variables defined in StreamTokenizer. Also, explain use of StreamTokenizer with the help of an example. *10*

(b) "Java is architecture neutral, secure and distributed programming language." Justify the statement. *5*

(c) What is private access specifier ? Explain with an example, how it is different from public and protected access specifiers. *5*

4. (a) What is finalize( ) method in Java. Give an example to demonstrate its use. 5

(b) Define package in Java. Explain the need of 'user created package' in Java. Also, explain the procedure of creating a package in Java. 5

(c) What is the difference between *throw* and *throws* keywords used in Java ? Explain with an example. 5

(d) Explain use of *super* keyword in Java, with an example. 5

5. (a) What is stateless protocol ? Is HTTP a stateless protocol ? What are the various ways through which you can maintain the 'State' while using HTTP ? 8

(b) Differentiate between TCP client and TCP server socket with the help of an example. 4

(c) What is RMI ? Explain advantages of using RMI. 3

(d) Briefly explain the life cycle of a servlet. 5

No. of Printed Pages : 3

MCS-024

12559

MCA (Revised)/BCA (Revised)

Term-End Examination

June, 2014

MCS-024 : OBJECT ORIENTED TECHNOLOGIES AND JAVA PROGRAMMING

Time : 3 hours *Maximum Marks : **100***

(Weightage 75%)

Note : *(i) Question No. **1** is **compulsory**.*

*(ii) Answer **any three** questions from the rest.*

(iii) Attempt all of a question together.

---

1. (a) What is object oriented programming ? How is it different from structured programming ? Explain advantages of using object oriented programming. **8**

(b) Explain why is Java called platform independent language ? **3**

(c) What is abstraction ? Explain why classes in OOP, known as abstract data types. **5**

(d) What is Java applet ? How is it different from Java application program ? **5**

(e) Explain any five interfaces of the Java IO package. **5**

(f) What is Exception ? Explain how exceptions are handled in Java. Write a Queue class that throws an exception when the Queue is full. Display exception message as "Queue is full". 9

(g) Explain the Java RMI architcuture with the help of a diagram. 5

2. (a) What is multithreading ? Explain the steps for creating thread in Java. Also, explain different predefined thread priorities in Java. 10

(b) Write a Java Program which takes two 3×3 matrices as input and find sum of the two matrices ? Define constructor for initializing matrix objects. 10

3. (a) Write a Java program that reads the contents of a text file and counts the number of white spaces, number of characters, number of words, number of semicolons in the contents. Print the output of program on standard output. 10

(b) Write a Java program to find factorial of a number. 5

(c) What is inheritance ? Explain with an example how is multiple inheritance achieved in Java ? 5

4. (a) Write a Java program to setup JDBC and execute an SQL statement on a database table 'BOOK', with the fields BOOK_ISBN, BOOK_AUTH, BOOK_PRICE, BOOK_PUBL. The SQL statement should find the BOOK_AUTH and BOOK_PRICE of the Book having ISBN = 235001. **10**

(b) What is Polymorphism ? Explain advantage of polymorphism with an example of managing different types of Bank Accounts. **10**

5. (a) What is Java Bean ? Explain its advantages. **5**

(b) What is String Class in Java ? Write a program to insert a substring in the middle of a string. (Do not use methods of string class) **5**

(c) What is layout manager ? Explain any two layouts in Java briefly. **5**

(d) Differentiate between overloading and overriding with the help of suitable example. **5**

No. of Printed Pages : 3

MCS-024

MCA (Revised)

Term-End Examination

December, 2013

MCS-024 : OBJECT ORIENTED TECHNOLOGIES AND JAVA PROGRAMMING

Time : 3 hours *Maximum Marks : 100*

Note : *Question No. 1 is **compulsory**. Attempt **any three** questions from the rest. Attempt all parts of questions together.*

---

1. (a) Bring out the difference between the statement and prepared statement classes in Java. **5**

(b) What is the JVM ? How does it helps to make Java machine independent ? **5**

(c) What is Java bean ? List its main features. **5**

(d) Draw a chart to show in a class, for each access specifier, which member (class, package, subclass or all) may NOT reference data or methods of that class ? **4**

(e) Explain when the transient and volatile modifiers are used ? **6**

(f) Describe the concept of method overriding, with an example. **5**

(g) Write a Java program to count the number of words and lines in a test file using the string Tokenizer class. **10**

2. (a) Write an applet that draws a coloured line from the last point on which the mouse was clicked to the location of the current mouse click. The colour of the line should be a parameter passed to the applet when invoking it. **10**

(b) Write a Java program to show that when more than one thread is waiting for an object's monitor, and the monitor wakes up all threads waiting on it, the highest priority thread that has been waiting is run first. **10**

3. (a) There are two files called file 1 and file 2 each containing some lines of text (such as names of people) in sorted order. Write a program to print total number of words in both the files. Print which file has larger number of words. **10**

(b) Arrange the following operators in descending order of precedence ?, &&, ~, instance of, + +, |, ||, % **4**

(c) (i) How can you prevent a class from being inherited ? **1**

(ii) Can an abstract class have a method that is not abstract ? **1**

(iii) Can a class reference the members of its superclass ? **1**

(iv) Is multiple inheritance allowed in Java ? Is multi-level inheritance allowed ? **2**

(v) Can you have a try block without a catch clause ? **1**

4. (a) In a two dimensional Java array, can each row have a different number of columns ? Write a Java program to initialize the contents of a 3 dimensional array of size $9 \times 6 \times 11$ such that each element is the sum of the cubes of its subscripts. For example, a $[5][4][8] = 125 + 64 + 512 = 701$ **10**

(b) Write a Java program to receive a UDP datagram on port 9701 and place the data received into a file. **10**

5. (a) Write a Java class that takes as input an integer array of numbers and prints out the sum of all the elements if there are 16 or less. If there are more elements its throws an exception which gives a message showing how many extra elements (more than 16) there are in the array. **10**

(b) What is an interface ? How is it declared and used ? How does it differ from an abstract class ? What is meant by implementing an interface ? **10**

No. of Printed Pages : 4

MCS-024

01354

MCA (Revised)

Term-End Examination

June, 2013

MCS-024 : OBJECT ORIENTED TECHNOLOGIES AND JAVA PROGRAMMING

Time : 3 hours *Maximum Marks : 100*

Note : *Question No. 1 is **compulsory**. Attempt **any three** questions from the rest. Attempt all parts of questions together.*

1. (a) Bring out the difference between the string and stringbuffer classes in Java. 5

   (b) Explain the concept of inheritance. Give a suitable example by declaring a class and showing two levels of inherited classes with the methods of each. 7

   (c) There are some classes that reside in a public package called my Package. Give three ways in which these can be accessed in the default unnamed package. If one of the classes in my Package has the same name as a class in the default package, how can it be referred to in the default package ? 4

   (d) What are the steps involved in creating a distributed application using Remote Method Invocation ? 6

(e) (i) How would you transfer control from the innermost loop of a thrice nested loop to the next statement outside the loops ? 1

(ii) What are the conditions necessary for a class to be serializable ? 1

(iii) Why is it not necessary to handle Runtime Exception ? 2

(f) When is the "this" keyword used and why ? Write a program fragment to illustrate. 4

(g) Write a Java program to read a text file and output the number of case insensitive occurrences of the sequence of letters "the" in it. 10

2. (a) What are Java controls and containers ? Write an applet that displays three labels and four non-exclusive checkboxes on the screen. 8

(b) Construct a programming problem where the switch statement cannot be used for taking decisions. Now write a Java program to solve the problem. 6

(c) Write a program to track sessions through hidden form fields. What is the disadvantage of this approach ? 6

3. (a) What is polymorphism ? Write a program to illustrate late binding in Java. 10

(b) "Experiment 1" is a text file. Write a Java program to read" Experiment 1" and copy the contents of "Experiment 1" to a new file "Experiment 2". Also count the number of words and lines copied. 10

4. (a) There is a database with a table emp_t that has employee_id, name, old_basic_salary and new_basic. Write a Java program to find all employees whose old basic salary is less than ₹ 20,000 and give them a 10% raise, while all others get a 6% raise. Update emp_t with the new basic salary of each employee. 12

(b) Suppose x in octal notation is $(567)_8$ and similarly $y = (-325)_8$. Write down the values of : 8

(i) $x<<y$ (ii) x/y

(iii) $x \& y$ (iv) $x>>3$

(v) $x >>> 3$ (vi) $x \wedge y$

(vii) $\sim x$ (viii) $\sim y$

5. (a) What is meant by thread priority ? Write a Java program that creates two threads-one with minimum and the other with maximum priority. The first prints the letters A to Z while the second prints the numbers 1 to 26. Which thread would you expect to finish first and why ? 8

(b) Write a Java program that finds : 8

(i) the area of a circle with that radius if 1 parameter is passed to it.

(ii) the area of a rectangle with those sides if 2 parameters are passed to it.

(iii) the surface area of a box if 3 parameters are passed to it.

(c) Explain how to compile and run a Java Servlet. 4

No. of Printed Pages : 3

MCS-024

MCA (Revised)

Term-End Examination

December, 2012

01585

MCS-024 : OBJECT ORIENTED TECHNOLOGIES AND JAVA PROGRAMMING

Time : 3 hours *Maximum Marks : **100***

Note : *Question **1** is **compulsory**. Attempt **any three** from the rest.*

---

1. (a) Why java is known as architectural neutral? **2**

(b) What is method overloading? Explain your answer with suitable example. **5**

(c) Explain static variable is used in java? **5**

(d) What are the functions of the dot(.) operator ? Explain with suitable example. **5**

(e) Differentiate between Inheritance and Abstract class ? **5**

(f) What is Interface in Java? Explain with an example how interfaces are created in Java. **6**

(g) What is multithreading ? Explain Different stages of a thread. **6**

(h) What is Garbage collection? Explain advantages and disadvantage of Garbage collection in Java. **6**

2. (a) Briefly explain role of following classes in Java Network programming 6

(i) Socket

(ii) Datagrams Packet

(b) Explain the difference between application and applet ? 4

(c) What is object oriented programming ? Explain basic concepts of object oriented programming, briefly. 10

3. (a) Explain the two types of exceptions available in java, with an example of each. 8

(b) Explain when should you extend the thread class for creating a thread. 4

(c) Write a program to read the contents of file into a character away and write it into another file. Get names of the file from the user through standard input. 8

4. (a) Can we import the same package class twice? Will the JUM load the package twice at runtime? Justify your answer. 5

(b) What are the characteristics of JDBC ? What are the various types of JDBC? Write a program to demonstrate how JDBC connection is established. 8

(c) WAP to reverse a linked list and then copy all elements of a linked list into another list. 7

5. (a) What is String Buffer? How it is different from String? Write a Java program which take your name as input and print it in upper case. 8

(b) What is RMI ? Explain RMI Architecture with suitable diagram. 7

(c) Why we use Java Beans in Java? Explain features of Java Beans. 5

No. of Printed Pages : 3

MCS-024

03337

MCA (Revised)

Term-End Examination

June, 2012

MCS-024 : OBJECT ORIENTED TECHNOLOGIES AND JAVA PROGRAMMING

Time : 3 hours *Maximum Marks : **100***

Note : *Question **1** is **compulsory**. Attempt **any three** from the rest.*

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1. (a) What is multithreading ? Explain how does it help Java in its performance ? **5**

(b) Differentiate between throw and throws ? **5**

(c) What is Inheritance ? Briefly explain importance of super keyword in Java ? **5**

(d) What are shift operators ? How many types of shift operators are available in Java ? **5**

(e) Briefly explain benefits of stream classes. **5**

(f) Explain URL rewriting with an example. **5**

(g) Explain how exception handling is done in Java, with the help of an example. **5**

(h) What is a package in Java ? Explain how package is created in Java. **5**

2. (a) Distinguish between the following terms with examples : 2.5x4=10

(i) Exception and Error

(ii) Method overloading and overriding

(iii) Final and Finally

(iv) Instance variables and class variables.

(b) What is constructor ? Explain constructor overloading in Java with an example. 10

3. (a) What is the common usage of serialization? 3

(b) What is the result of compiling and running the following program ? 5

```
public class test
{
Public static void main (string args[ ])
{
int i = -1;
i = i>>1 ;
system.out. println (i) ;
}
}
```

(c) Explain use of keyword *this* with the help of program ? 3

(d) Create a class with in this package "Amount Inwords" to convert amount into words. (Consider amount not to be more than 100000) 9

4. (a) What is URL ? Explain two constructor for URL, in Java.net package. Explain how you may connect to a URL in Java. 8

(b) Write an applet that draws circle, a line, and a polygon inside the applet's visible area. 5

(c) Compare the different layout managers in brief. 7

5. (a) Write a serulet program that fetches all data from client and stores it in a database successfully. 8

(b) What do you mean by an event ? Explain different components of an event. 6

(c) What are the different types of AWT components ? How are these components added to the container. 6

No. of Printed Pages : 2

MCS-024

10152

MCA (Revised)

Term-End Examination

June, 2011

MCS-024 : OBJECT ORIENTED TECHNOLOGIES AND JAVA PROGRAMMING

Time : 3 hours *Maximum Marks : 100*

Note : *Q. No. 1 is **compulsory**, Attempt **any** 3 questions from the rest. Attempt all parts of questions together.*

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1. (a) Explain, with suitable examples, the advantage of object oriented language over structured programming language. **7**

   (b) Differentiate between method overloading and method overriding with an example. **5**

   (c) What is the use of class path ? How it helps in the execution of a java program ? **4**

   (d) Explain the advantages and disadvantages of garbage collection. **5**

   (e) What is the use of Interface ? How can you define and implement it using a program ? **6**

   (f) Differentiate checked and unchecked exception. **3**

   (g) Write a program in JAVA to convert given Full Name eg. "Ajay Singh" into First Name i.e "Ajay" and Last Name i.e "Singh". Also sort the given list of names in ascending order of First Name ? **10**

2. (a) How does java implements the model of interprocess synchronization using Threads ? Explain with the help of a program. 10

(b) Write a program in Java which reads the content of a given file and writes it to the console. 10

3. (a) What is layout manager ? Which is the default layout manager ? Differentiate among the way of managing components by 5 different layouts. 10

(b) Write a program for reading from and writing to a URL connection. 10

4. (a) Explain advantage of exception. How exception sub classes are created ? Create your own exception subclass "My Ex class" which takes an string as input and throws a message "string too large" when the size of string is more than ten characters. 10

(b) Differentiate between Datagram and stream communication. 5

(c) Explain two uses of "final" keyword with the help of example. 5

5. (a) Write the process of implementing cookies. What is the role of session in handling a cookie. 10

(b) What is Inheritance ? Explain its advantages. Also explain with example how a subclass is derived from a super class in Java. 10

No. of Printed Pages : 3

MCS-024

08086

MCA (Revised)

Term-End Examination

December, 2010

MCS-024 : OBJECT ORIENTED TECHNOLOGIES AND JAVA PROGRAMMING

Time : 3 hours *Maximum Marks : 100*

Note : *Question. No. 1 is* ***compulsory****. Attempt* ***any three*** *questions from the rest.*

---

1. (a) Write a Java program to read the contents of binary file as integers and print out their values on the standard output. **10**

   (b) What are anonymous classes ? Explain with a code fragment. **5**

   (c) What is the purpose of serialization of an object ? How does the volatile modifier to a data type affect serialization ? **5**

   (d) Justify "Java is a secure" and "architectural neutral" Language. **5**

   (e) Write a program to generate a fibonaaci series. **5**

   (f) What is a synchronized block ? **5**

   (g) What is the difference between the notify ( ) and notify All ( ) methods ? **5**

2. (a) Explain the purpose of Inet Address class and list three of its methods. 5

(b) Explain the difference between 5

(a & b) and (a & & b).

where a and b are logical expressions.

(c) Explain with an example the use of the ternary operator. 5

(d) Java is architecture neutral, secure and distributed language. Justify. 5

3. (a) Develop an applet to display a solid circle that fits perfectly inside the outline of a square. 10

(b) What is the difference between a process and a thread ? Also mention **any three** methods of thread class and describe briefly. 8

(c) What is grid layout ? 2

4. (a) Write a Java applet to present pull down list of the seven colours (violet, indigo, blue, green, yellow, orange red) of a rainbow and allow the user to select one of the colours. Print out the index number of the selected choice. 10

(b) What are the principles of event delegation model ? What are the sources of event and event listener interface ? 7

(c) Explain briefly the concept of RMI. 3

5. (a) Write a Java program that draws a line, a triangle or a circle depending on whether 2, 3 or 1 point parameters are given to it : 8

Line between the two points

Triangle joining the three points

Circle of default radius at that point

(b) What are checked and unchecked exceptions ? Explain with example. 7

(c) What is wrong with the following program fragment ? Make the needed changes to correct it. 5

```
double d ;
d =4.75 ;
switch (d)
{
case 0 :
case 1 :
case 2 : { Println ("small") ;
           break} ;
case 3 :
case 4 :
case 5 : { println ("medium") ;
           break } ;
default :
           println ("large") ;
}
```

No. of Printed Pages : 4

MCS-024

01042

MCA (Revised)

Term-End Examination

June, 2010

MCS-024 : OBJECT ORIENTED TECHNOLOGIES AND JAVA PROGRAMMING

Time : 3 hours *Maximum Marks : 100*

Note : *Q. No. **1** is **compulsory**. Attempt **any three** questions from the rest.*

---

1. (a) Consider a class that stores a Bank account holder's account number, ATM card number, account balance and ATM PIN. Write a program to store the data onto a disk file, except for the account balance and ATM PIN. Use serialization and transient variables. **6**

(b) What makes compiled JAVA programs platform independent ? Explain. **4**

(c) Explain the significance of PATH and CLASS PATH. **5**

(d) Is this a complete and legal comment ? **2**

/ * // */

Justify .

(e) Are inner class secure and useful ? **5**

(f) What are the two fundamental mechanisms for building a new classes of a existing class ? **5**

(g) The members of which classes can access members of a class in a particular package ? 5

(h) Explain the significance of each argument in the function main (string args [ ]). 5

(i) Why cannot the "abstract" method modifier be used together with the "native" modifier ? 3

2. (a) Differentiate the following and support with example : 2.5x4=10

(i) Final and static member

(ii) Inheritance and Aggregation

(iii) Abstract class and Interface

(iv) String and String Buffer

(b) Write a program to implement the following operations on a string : 3.5x2=7

(i) Convert to uppercase

(ii) Convert to lowercase

(c) What are user - defined exceptions ? 3

3. (a) What is the difference between checked and unchecked exceptions ? 3

(b) Create a class account with the following data members : 5

(i) account id

(ii) name

(iii) current balance

and implement the member functions :

(i) Withdraw

(ii) deposit

(c) How multiple inheritance is achieved in Java ? Explain with an example. 3

(d) Write an applet that prints "Here" at the current cursor position whenever the mouse left button is clicked. 9

4. (a) Write short notes on following :

(i) Interthread communication 3

(ii) Event Handling in Java 3

(b) What is a layout managers ? Explain layouts with example. 6

(c) Write a Java program to set up JDBC and execute the following SQL statement on a database table employee-t with the fields emp-id, emp name, emp-department, emp-basic "SELECT * FROM employee-t where emp-basic < 10000; 8

5. (a) Write a program to sort the list given as command line arguments using bubble sort. **10**

(b) What is Java Bean ? What are its advantages ? **5**

(c) What is Introspection in Java Beans ? Illustrate with a code fragment. **5**